

Eliminate the need for seal gas boosters with our Seal Protect solution

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Seal Protect can be installed on new compressor packages with dry gas seals; or it can be retrofitted into existing units, providing an upgrade option for added reliability

Seal Protect is a reliable and cost-effective sealing mechanism that enables operators to eliminate the need for a seal gas booster by protecting the dry gas seals from process gas contamination at slow speeds, or when the compressor has stopped in a pressurized standby condition.

Operators frequently identify the gas seal panel and booster compressor as some of the most failure-prone components in compressor packages. Seal Protect can be installed as a low-cost alternative to conventional seal gas boosters, simplifying the system design and enhancing the overall reliability. It can be a valuable addition to any dry gas seal system in an oil and gas application by providing an added layer of protection against contamination, reducing the risk of downtime and associated expenses from maintenance and lost production.

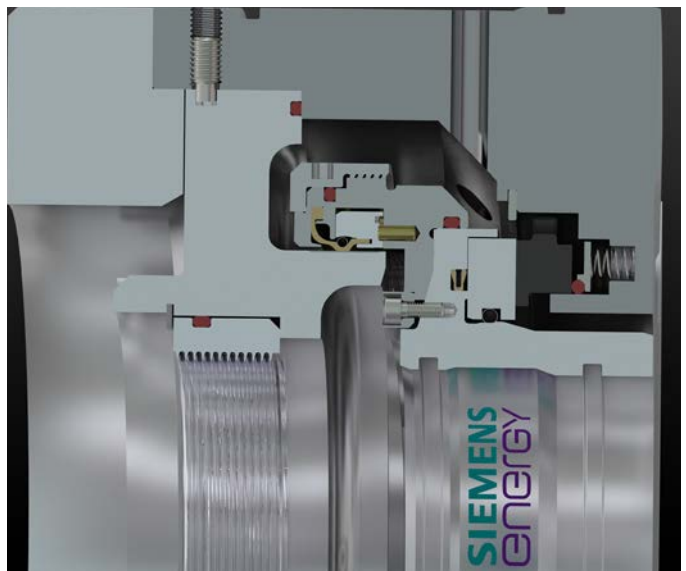
The design provides added protection

This unique sealing solution sits between the dry gas seal and process side seal.

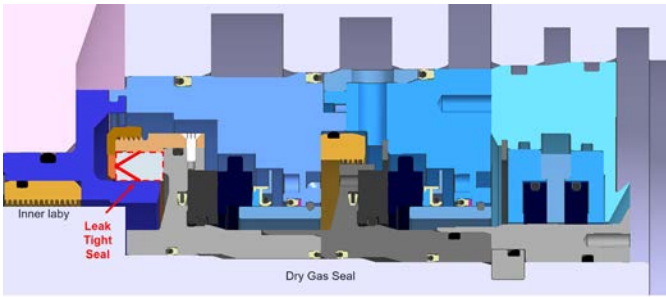
Seal Protect is wide open during normal operation, so that it is not subjected to any wear from high velocity rotation. Here, the dry gas seal is protected from contamination by the flow under inner laby, as usual.

Under static conditions and low speed, Seal Protect is closed and prevents contamination of the dry gas seal from the inboard side.

Unfiltered process gas is blocked by Seal Protect and cannot enter the dry gas seal. Consequently, the dry gas seal is exclusively exposed to clean, dry process gas.



Seal Protect coupled with Siemens Energy dry gas seal system.



Seal Protect sits between the dry gas seal and the process side seal. With the elimination of the booster, this new design enables lower emissions.

Benefits

- Increased reliability versus reciprocating seal gas boosters - Simplifying the system by reducing the number of components and connections simplifying the system decreases the risk of leaks or failure and enhances the overall system reliability.
- Reduced cost – CAPEX is lower than centrifugal booster compressors (although costs vary depending on the application). OPEX is also reduced as Seal Protect does not require additional electrical power, or air consumption and needs less heating. The required maintenance is also considerably less compared to reciprocating seal gas boosters.
- Reduced footprint - Often, the seal gas booster is incorporated within a pre-conditioning skid, separate from the gas seal panel. By installing Seal Protect, it may be possible to remove the skid, potentially minimizing the overall package footprint.
- Lower emissions - Utilizing Seal Protect instead of an electric-driven booster compressor results in decreased power consumption and related emissions from the compressor package.

Seal Protect serves as a low-cost alternative to boosters and is compatible with most new compressor packages equipped with dry gas seals. Additionally, it can be retrofitted into existing units, providing an upgrade option for those seeking improved reliability.

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